Phase 1

Dataset & Research Proposal

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Section 1: Title

- "Genre and Platform as Predictors of North American Video Games Sales"

Section 2: Research Question

- Can a game's genre and platform be used to predict its sales performance in the North American Market, among platforms with a substantial number of released titles? Given the known regional differences in gaming preferences; for example, action and shooter games have historically performed well in the Western markets. We expect genre and platform to be meaningful predictors of the sales in the North American Markets.
- One row in our database represents a single video title released on a specific platform, along with its genre, publisher, release data, critic score and global/regional sales figures.

Section 3: Dataset Information

- Dataset Source: https://www.kaggle.com/datasets/lamskdna/video-games-sales/data
- Collection Method: VGChartz
- License: Apache 2.0, Kaggle
- Number of Rows: 64016.
- Number of Columns: 12.
- Data Types: Numeric, date, text.
- Known Limitations: Dataset collected through scraping a website

Section 4: Planned EDA Questions

1. What percentage of rows have missing or ambiguous values?
2. Do average na_sales(mil) diverge heavily per genre?
3. Does a single console's na_sales(mil) outperform others by a substantial amount?
4. How many distinct consoles are represented in the dataset, and what are the amounts of released titles per platform?

Section 5: Anticipated Challenges

1. With a dataset of this size, we expect that some rows will have missing or true NaN values, resulting in a judgement call on eliminating that row and potentially affecting the genre/platform sales average if handled incorrectly.
2. There is an unfair representation in the genre column, there exists a Misc. category which include more than 5000 titles while boardGame and sandbox titles equate to less than a hundred. In order to allow for fair representation of genres we will set a minimum number of titles per genre to include in our analysis.
3. Moreover, there is another imbalance in our console column. PC holds nearly 10,000 titles while outdated platforms only have several titles to their name. To alleviate the possibly of skewed data with numerous underrepresented consoles we will once again set a minimum title count to console to reflect the primary time range of our dataset.